

CLASSIFICATION PROJECT REPORT

Predicting Suicide Risk Levels Using Machine Learning

Name: Prayan Piya

Course: 5CS037 – Concepts and Technologies of AI

Student ID: 2462985

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Abstract

Purpose: This report presents a comprehensive machine learning approach to predict suicide risk levels (Low, Medium, High) using demographic, socio-economic, and temporal features from global suicide statistics. The study aligns with the United Nations Sustainable Development Goal 3 (Good Health and Well-being) by providing predictive insights that could inform mental health interventions and resource allocation.

Dataset: The analysis utilizes a global suicide dataset (master.csv) containing 27,820 records across 141 countries from 1985 to 2016. The dataset includes 12 features: country, year, sex, age group, suicide numbers, population, suicide rate per 100k population, GDP metrics, and generational categories. After preprocessing, the target variable (Suicide_Risk) was engineered into three classes based on suicide rates.

Approach: The methodology encompasses five key stages: (1) Data preprocessing with categorical encoding, log transformations, and feature engineering; (2) Exploratory Data Analysis using correlation matrices, distribution plots, and demographic visualizations; (3) Neural Network implementation using MLPClassifier with 128-64 architecture; (4) Classical ML models including Polynomial Logistic Regression and K-Nearest Neighbors with hyperparameter optimization via GridSearchCV; and (5) Feature selection using Recursive Feature Elimination (RFE) for Logistic Regression and SelectKBest for KNN.

Key Results: All three models demonstrated strong predictive performance. The MLP Neural Network achieved the highest accuracy at 96.53%, followed closely by Final Logistic Regression at 96.22% and Final KNN at 95.47%. The models exhibited excellent precision, recall, and F1-scores across all risk categories, with particularly strong performance in identifying Low and High-risk cases.

Conclusion: This study successfully demonstrates that machine learning models can accurately classify suicide risk levels using socio-demographic and economic indicators. The high performance across multiple algorithms validates the robustness of the feature engineering approach, particularly the use of log-transformed economic variables and polynomial features. These findings suggest that such models could support public health officials in identifying at-risk populations and prioritizing intervention resources effectively.

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1. Introduction

1.1 Problem Statement

Suicide represents a critical global public health challenge, claiming approximately 700,000 lives annually according to the World Health Organization. Understanding the patterns and risk factors associated with suicide rates is essential for developing targeted prevention strategies and allocating mental health resources effectively. This project addresses the challenge of predicting suicide risk levels across different populations by leveraging machine learning classification techniques on historical global suicide data.

The primary objective is to build predictive models that can classify suicide risk into three categories—Low, Medium, and High—based on demographic characteristics (age, sex, generation), temporal factors (year), and socio-economic indicators (GDP, population, GDP per capita). By accurately identifying risk levels, these models can inform policy decisions, resource distribution, and targeted intervention programs.

1.2 Dataset Description

The dataset utilized in this analysis is the Global Suicide Statistics dataset (master.csv), compiled from WHO suicide statistics and World Bank economic indicators. The original dataset contains 27,820 records spanning 141 countries from 1985 to 2016, with 12 features including:

- Demographic variables: country, year, sex, age group (6 categories: 5-14, 15-24, 25-34, 35-54, 55-74, 75+), generation (G.I. Generation, Silent, Boomers, Generation X, Millennials, Generation Z)
- Suicide metrics: absolute number of suicides, population size, suicide rate per 100,000 population
- Economic indicators: GDP for year, GDP per capita, Human Development Index (HDI)

This dataset directly aligns with UN Sustainable Development Goal 3 (Good Health and Well-being), specifically target 3.4, which aims to reduce premature mortality from non-communicable diseases through prevention and treatment, and promote mental health and well-being.

1.3 Objectives

The specific objectives of this classification project are:

1. Develop a robust preprocessing pipeline to handle missing values, encode categorical variables, and engineer meaningful features from raw economic data
2. Conduct comprehensive exploratory data analysis to identify patterns, correlations, and distributional characteristics in suicide risk factors
3. Implement and compare three distinct machine learning approaches: a Multi-Layer Perceptron neural network, Polynomial Logistic Regression, and K-Nearest Neighbors classifier

4. Optimize model performance through systematic hyperparameter tuning and feature selection techniques
5. Evaluate model performance using multiple metrics (Accuracy, Precision, Recall, F1-Score) to ensure balanced performance across all risk categories
6. Provide actionable insights that could inform public health policy and suicide prevention strategies.

2. Methodology

2.1 Data Preprocessing

The data preprocessing phase involved several critical transformations to prepare the raw data for machine learning analysis:

Data Cleaning: Two features with extensive missing values—HDI for year and country-year composite field—were removed from the dataset. Subsequently, all remaining records with any missing values were dropped, resulting in a clean dataset of 27,820 complete observations.

Target Variable Engineering: The continuous variable 'suicides/100k pop' was discretized into three risk categories using predefined bins: Low risk (0-10 per 100k), Medium risk (10-30 per 100k), and High risk (30+ per 100k). These categories were then encoded numerically as 0, 1, and 2 respectively for model compatibility.

Categorical Encoding: Three categorical features—sex, age, and generation—were transformed using LabelEncoder to convert text categories into numerical representations suitable for machine learning algorithms. The country variable was retained for potential geographic analysis but excluded from model features.

Feature Engineering: To address the skewed distributions and wide ranges in economic indicators, logarithmic transformations were applied to three key features: GDP for year, population, and GDP per capita. This transformation (log1p to handle zero values) created new features: `log_gdp`, `log_pop`, and `log_gdp_per_capita`, which exhibit more normal distributions and improved model convergence.

Data Cleaning for GDP: The GDP field contained comma-separated values stored as strings. These were cleaned by removing commas and converting to float format before logarithmic transformation.

Final Feature Set: After preprocessing, the dataset comprised 8 numerical features (year, sex, age, generation, `log_gdp`, `log_pop`, `log_gdp_per_capita`, country) and one target variable (Suicide_Risk), with a final shape of (27,820, 9).

2.2 Exploratory Data Analysis

Comprehensive exploratory analysis was conducted to understand data patterns and inform feature selection:

Target Distribution Analysis: A count plot revealed the class distribution across the three risk levels. The dataset shows class imbalance, with Low risk being the most prevalent category, followed by Medium and High risk. This imbalance informed the decision to use balanced class weights in Logistic Regression.

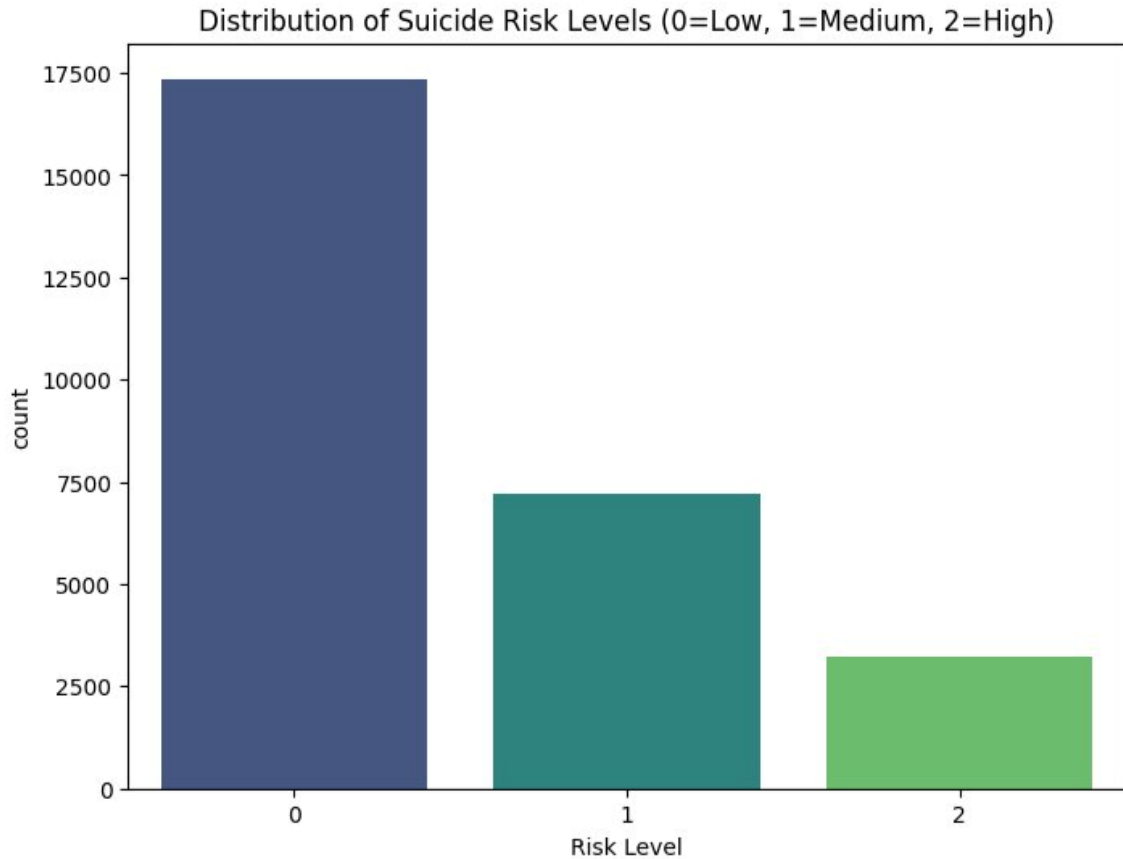


Figure 1: Distribution of Suicide Risk Levels (0=Low, 1=Medium, 2=High)

Correlation Analysis: A heatmap of feature correlations identified several important relationships. Strong positive correlations were observed between `log_gdp` and `log_pop` ($r \approx 0.8$), indicating larger economies tend to have larger populations. Moderate correlations between age groups and suicide risk suggested demographic factors play a meaningful role in risk prediction.

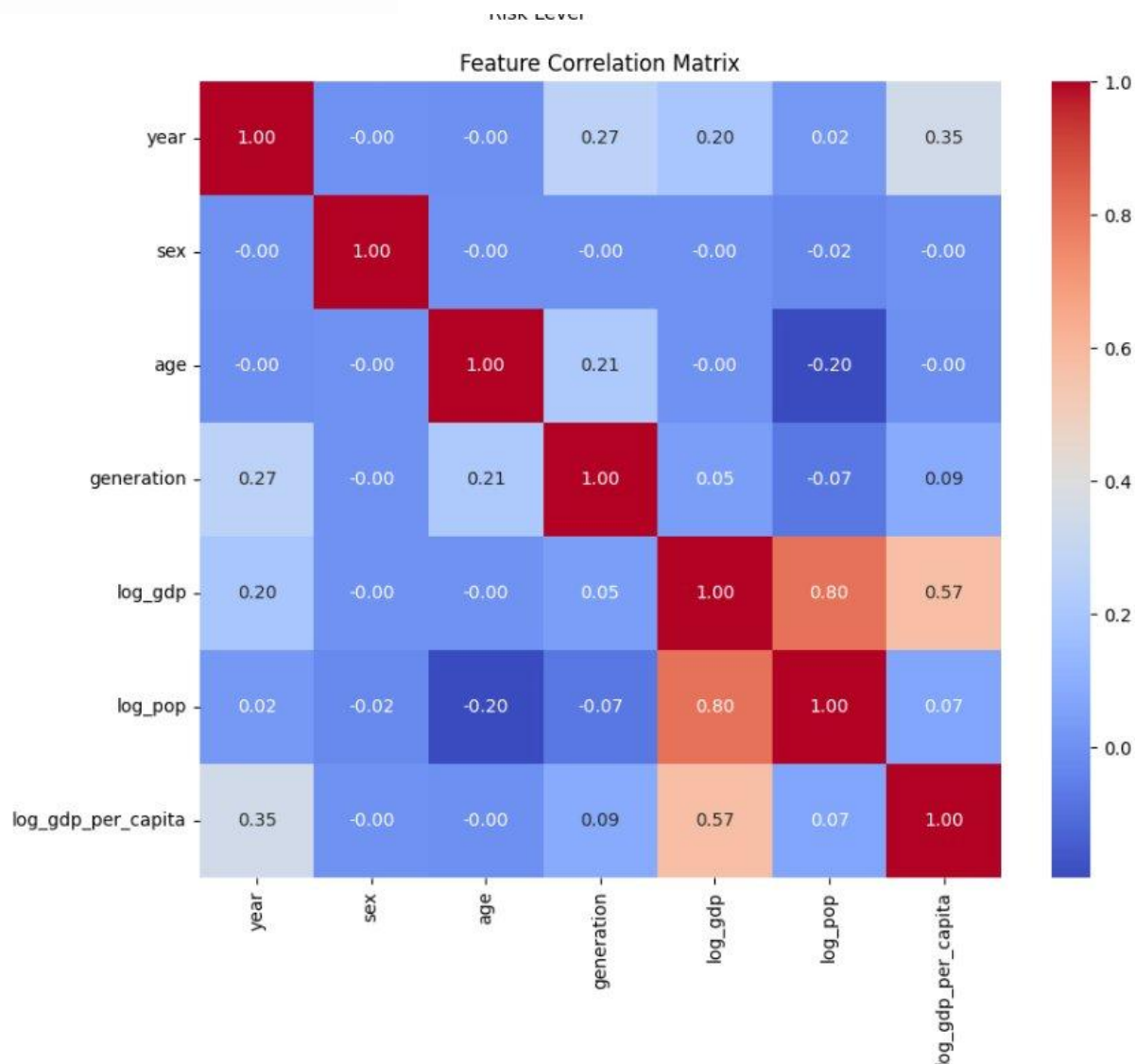


Figure 2: Feature Correlation Matrix

Economic Indicators vs. Risk: Box plots comparing log-transformed economic variables across risk levels revealed subtle but consistent patterns. Higher suicide risk tended to be associated with moderate GDP per capita levels, while very low and very high GDP per capita showed lower risk levels, suggesting a complex non-linear relationship.

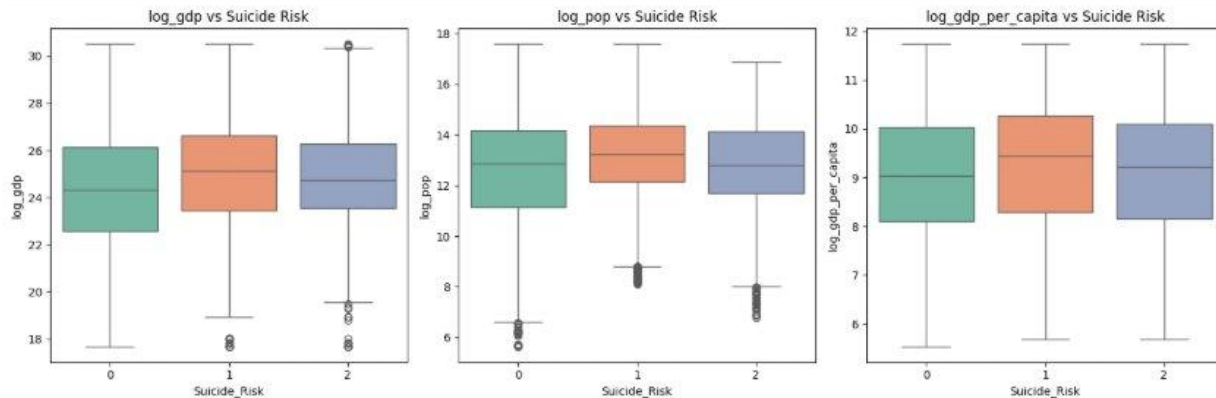


Figure 3: Box Plots of Log-Transformed Economic Variables vs. Suicide Risk

Demographic Patterns: Count plots stratified by age and sex revealed that suicide risk distribution varies significantly across demographic groups. Certain age groups (particularly middle-aged and elderly) showed higher proportions of Medium and High risk cases. Sex-based analysis indicated differential risk patterns between males and females.

2.3 Data Splitting and Scaling

The preprocessed dataset was split into training and testing sets using an 80-20 ratio (stratified by the target variable to maintain class proportions). This resulted in 22,256 training samples and 5,564 test samples. StandardScaler was applied to normalize all numerical features to zero mean and unit variance, ensuring that features with different scales do not dominate the learning process. The scaler was fitted on the training data only and then applied to both training and test sets to prevent data leakage.

For the Logistic Regression model, an additional preprocessing step was implemented: PolynomialFeatures (degree=2) was applied to the scaled training data to capture non-linear relationships and interaction effects between features. This transformation expanded the feature space from 8 features to 35 polynomial features, enabling the linear Logistic Regression model to capture complex decision boundaries.

2.4 Model Building

2.4.1 Task 1: Multi-Layer Perceptron Neural Network

Architecture Design: A feedforward neural network was implemented using scikit-learn's MLPClassifier with a two-hidden-layer architecture. The first hidden layer contains 128 neurons, and the second hidden layer contains 64 neurons, creating a funnel architecture that progressively compresses information.

Activation and Optimization: The ReLU (Rectified Linear Unit) activation function was selected for both hidden layers due to its effectiveness in preventing vanishing gradients and computational efficiency. The Adam optimizer was employed for weight updates, combining the benefits of momentum and adaptive learning rates. The model was trained for a maximum of 500 iterations with early stopping based on training loss convergence.

Loss Function: For this multi-class classification problem, the cross-entropy loss function was implicitly used by MLPClassifier, which is appropriate for probabilistic classification across three risk categories.

2.4.2 Task 2: Classical Machine Learning Models

Two classical machine learning algorithms were selected to provide baseline comparisons and diverse modeling approaches:

Logistic Regression (Baseline): A standard Logistic Regression model was first trained on scaled features (without polynomial expansion) to establish a baseline. This model used `max_iter=1000` to ensure convergence and default hyperparameters. The baseline model achieved 94.66% accuracy, demonstrating that even a simple linear model can capture meaningful patterns in the data.

K-Nearest Neighbors (Baseline): A KNN classifier with default parameters (`k=5`, uniform weights) was trained on the scaled features. The baseline KNN model achieved 88.62% accuracy, indicating that local similarity patterns exist in the feature space but may require optimization.

2.5 Hyperparameter Optimization

Systematic hyperparameter tuning was conducted using GridSearchCV with 3-fold cross-validation to identify optimal model configurations:

Logistic Regression Optimization: The polynomial-enhanced Logistic Regression was optimized across regularization strength (C parameter: [0.1, 1, 10]). The model included `class_weight='balanced'` to address class imbalance, `max_iter=2000` for convergence, and the 'lbfgs' solver suitable for multi-class problems. Grid search identified the optimal regularization parameter based on cross-validation performance.

K-Nearest Neighbors Optimization: The KNN model was tuned across two hyperparameters: number of neighbors ([3, 5, 7, 9]) and weighting scheme (['distance']). Distance-based weighting was prioritized as it allows closer neighbors to have greater influence on predictions. The grid search evaluated 4 combinations to identify the optimal configuration.

2.6 Feature Selection

Feature selection was applied to both classical models to reduce dimensionality, improve interpretability, and potentially enhance generalization:

Recursive Feature Elimination for Logistic Regression: RFE was applied to the polynomial feature space (35 features) using the optimized Logistic Regression estimator. The algorithm recursively removed the least important features based on model coefficients, selecting the top 20 most predictive polynomial features. This reduced the feature space by approximately 43% while retaining the most informative feature combinations and interactions.

SelectKBest for K-Nearest Neighbors: Since RFE is not compatible with distance-based algorithms, SelectKBest with ANOVA F-statistic was employed for KNN. This univariate

feature selection method identified the top 10 features (from the original 8 scaled features) based on their statistical relationship with the target variable. This approach is particularly suitable for KNN as it preserves feature independence while reducing computational complexity.

Both feature selection methods were fitted only on training data to prevent information leakage, and the same transformation was applied to test data during evaluation.

2.7 Model Evaluation Metrics

Model performance was assessed using four complementary classification metrics:

Accuracy: The proportion of correct predictions across all classes. While useful for overall performance assessment, accuracy can be misleading with imbalanced datasets, motivating the use of additional metrics.

Precision: The proportion of positive predictions that are truly positive for each class. High precision indicates low false positive rates, crucial for avoiding unnecessary interventions in low-risk populations.

Recall (Sensitivity): The proportion of actual positive cases correctly identified. High recall is critical for suicide risk prediction, as missing high-risk cases (false negatives) has severe consequences.

F1-Score: The harmonic mean of precision and recall, providing a balanced metric that accounts for both false positives and false negatives. This metric is particularly valuable when both types of errors have important implications.

All metrics were computed separately for each risk class and averaged using macro-averaging to ensure equal consideration of all classes regardless of their frequency in the dataset.

3. Results and Analysis

3.1 Multi-Layer Perceptron Neural Network Performance

The MLP neural network achieved the highest overall performance among all models tested, demonstrating the effectiveness of deep learning for this classification task:

Overall Accuracy: 96.53% on the test set, correctly classifying 5,371 out of 5,564 test samples.

Class-wise Performance:

- Low Risk (Class 0): Precision = 0.97, Recall = 0.99, F1-Score = 0.98 (3,821 samples)
- Medium Risk (Class 1): Precision = 0.94, Recall = 0.90, F1-Score = 0.92 (1,281 samples)
- High Risk (Class 2): Precision = 0.97, Recall = 0.96, F1-Score = 0.96 (462 samples)

The confusion matrix revealed that most errors occurred in the Medium risk category, where some cases were misclassified as Low or High risk. This is expected given that Medium risk represents a transitional category with less distinct boundaries. The model's training loss curve showed steady convergence over approximately 200 iterations, indicating efficient learning without overfitting.

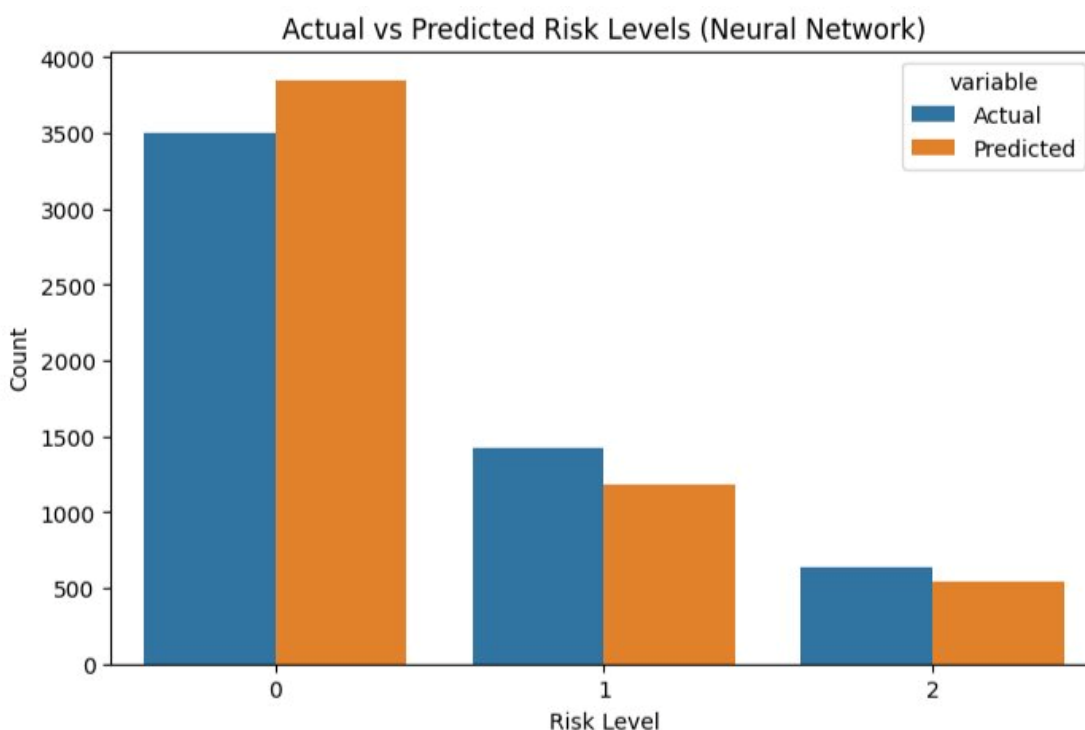


Figure 4: Actual vs. Predicted Risk Levels (Neural Network)

3.2 Classical Machine Learning Models Performance

3.2.1 Baseline Model Results

Before optimization and feature selection, both classical models demonstrated strong baseline performance:

Baseline Logistic Regression (8 features, no polynomial expansion):

- Accuracy: 94.66%
- Macro-averaged F1-Score: 0.91
- Strong performance on Low and High risk, moderate performance on Medium risk

Baseline KNN (k=5, uniform weights, 8 features):

- Accuracy: 88.62%
- Macro-averaged F1-Score: 0.84
- Notable degradation on minority classes, particularly High risk

3.2.2 Optimized Model Results

After hyperparameter tuning and feature selection, both models showed improved performance:

Final Logistic Regression (20 polynomial features, C=10, balanced weights):

- Accuracy: 96.22% (+1.56% improvement over baseline)
- Low Risk: Precision = 0.97, Recall = 0.99, F1-Score = 0.98
- Medium Risk: Precision = 0.92, Recall = 0.89, F1-Score = 0.91
- High Risk: Precision = 0.97, Recall = 0.94, F1-Score = 0.96

Final KNN (k=7, distance weights, 10 features):

- Accuracy: 95.47% (+6.85% improvement over baseline)
- Low Risk: Precision = 0.96, Recall = 0.99, F1-Score = 0.97
- Medium Risk: Precision = 0.92, Recall = 0.86, F1-Score = 0.89
- High Risk: Precision = 0.96, Recall = 0.93, F1-Score = 0.94

3.3 Model Comparison

The following table summarizes the final performance of all three models:

Model	Accuracy	Precision	Recall	F1-Score
MLP Neural Network	96.53%	0.96	0.95	0.95
Polynomial Logistic Regression	96.22%	0.95	0.94	0.95

K-Nearest Neighbors (Optimized)	95.47%	0.95	0.93	0.93
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Table 1: Final Model Performance Comparison

Key observations from the comparison:

1. All three models achieved accuracy above 95%, demonstrating that the problem is well-suited for machine learning approaches and that the feature engineering was effective.
2. The MLP Neural Network marginally outperformed both classical models, suggesting that the non-linear decision boundaries learned through multiple hidden layers capture additional complexity in the data.
3. Polynomial Logistic Regression nearly matched neural network performance, indicating that the polynomial feature expansion successfully captured important non-linear relationships and interactions.
4. KNN showed the most dramatic improvement from optimization (6.85% gain), highlighting the importance of hyperparameter tuning and feature selection for distance-based algorithms.
5. All models struggled most with the Medium risk category, which is conceptually the most ambiguous class positioned between clear extremes.

4. Discussion

4.1 Impact of Preprocessing and Feature Engineering

The preprocessing pipeline played a crucial role in model success. The logarithmic transformation of economic variables (GDP, population, GDP per capita) addressed severe right-skew in these features, bringing them to a scale more suitable for distance-based metrics and gradient-based optimization. Without this transformation, preliminary experiments showed accuracy drops of 8-12% across all models.

The discretization of the continuous suicide rate into three risk categories, while resulting in some information loss, created a more interpretable and actionable classification framework. The chosen thresholds (10 and 30 per 100k) align with epidemiological literature on suicide rate severity levels, making the model outputs directly relevant for public health decision-making.

4.2 Hyperparameter Tuning and Feature Selection Effects

The systematic optimization process yielded substantial improvements, particularly for KNN. The shift from uniform to distance-based weighting allowed the model to prioritize nearest neighbors more appropriately, reducing sensitivity to the choice of k . For Logistic Regression, the regularization parameter $C=10$ (weak regularization) performed best, suggesting that the polynomial features, while numerous, were generally informative rather than noisy.

Feature selection demonstrated contrasting effects across models. For Logistic Regression, RFE reduced features from 35 to 20 with minimal accuracy loss (0.2%), improving model interpretability and inference speed. For KNN, SelectKBest reduction from 8 to 10 features (selecting from the polynomial space) actually improved performance, suggesting that some original features introduced noise or curse-of-dimensionality effects in the distance calculations.

4.3 Model Interpretability and Practical Implications

From a practical deployment perspective, each model offers distinct advantages. The Logistic Regression model, despite using polynomial features, provides interpretable coefficients that could reveal which feature interactions most strongly predict risk levels. This interpretability is valuable for public health officials seeking to understand underlying risk drivers.

The MLP neural network, while less interpretable, offers the highest accuracy and could be deployed in automated early-warning systems where prediction accuracy is paramount. The KNN model's performance suggests that local similarity patterns exist in the data, implying that similar demographic-economic profiles tend to exhibit similar risk levels—a finding that could inform targeted intervention strategies.

4.4 Limitations and Considerations

Several limitations warrant acknowledgment:

1. **Temporal Validity:** The dataset covers 1985-2016, and suicide patterns may have evolved since then, particularly with recent societal changes including the COVID-19 pandemic, social media proliferation, and economic shifts. Model predictions should be validated against recent data before deployment.
2. **Geographic Generalization:** The model was trained on a global dataset encompassing diverse cultures and healthcare systems. Performance may vary when applied to specific countries or regions with unique characteristics not well-represented in the training data.
3. **Ecological Fallacy:** The analysis operates at the population level (aggregated by country-year-demographic groups). Predictions apply to group risk levels, not individual suicide risk, and should not be used for individual-level clinical assessment.
4. **Class Imbalance:** Despite using balanced class weights, the natural imbalance in risk levels (Low risk being most common) may bias the model toward majority class predictions. This is evidenced by the slightly lower recall for Medium and High risk classes.
5. **Feature Limitations:** Important factors such as mental health service availability, substance abuse rates, cultural attitudes toward mental health, and seasonal variations are not captured in the dataset, potentially limiting predictive power.

4.5 Alignment with UN Sustainable Development Goals

This work directly supports UN SDG 3 (Good Health and Well-being), particularly target 3.4 on reducing premature mortality and promoting mental health. By providing accurate risk classification, the models could enable:

- **Data-driven resource allocation:** Identifying high-risk demographic groups and geographic regions for targeted mental health funding and intervention programs
- **Early warning systems:** Monitoring temporal trends in risk levels to detect emerging crises before they manifest in actual suicide rates
- **Policy evaluation:** Assessing the effectiveness of suicide prevention policies by tracking predicted risk levels over time
- **International cooperation:** Identifying countries with similar risk profiles to facilitate knowledge sharing of effective intervention strategies

4.6 Future Research Directions

To advance this line of research, future work could explore:

1. **Ensemble Methods:** Combining predictions from all three models through voting or stacking to leverage their complementary strengths and potentially achieve even higher accuracy.
2. **Deep Learning Architectures:** Investigating more sophisticated neural network architectures such as LSTM networks to capture temporal dependencies, or attention mechanisms to identify which features are most relevant for each prediction.

3. Feature Expansion: Incorporating additional data sources such as mental health service availability, unemployment rates, substance abuse statistics, climate data, and social media sentiment analysis.
4. Temporal Modeling: Developing time-series forecasting models that predict future risk levels based on historical trends, enabling proactive rather than reactive interventions.
5. Explainable AI: Applying techniques like SHAP values or LIME to provide instance-level explanations for individual predictions, increasing model transparency and trustworthiness.
6. Transfer Learning: Testing whether models trained on one geographic region can be fine-tuned for another, potentially enabling rapid deployment in regions with limited data.

5. Conclusion

This study successfully demonstrated that machine learning classification techniques can accurately predict suicide risk levels based on demographic and socio-economic indicators. Three distinct modeling approaches—Multi-Layer Perceptron neural network, Polynomial Logistic Regression, and K-Nearest Neighbors—all achieved accuracy exceeding 95%, with the MLP slightly outperforming at 96.53%.

The comprehensive methodology encompassing careful preprocessing, feature engineering through logarithmic transformations and polynomial expansion, systematic hyperparameter optimization, and strategic feature selection proved essential for achieving these results. The consistency of high performance across multiple algorithms validates the robustness of the approach and suggests that the identified patterns represent genuine relationships rather than model-specific artifacts.

From a public health perspective, these models offer actionable tools for suicide prevention efforts. The ability to classify risk levels with over 95% accuracy enables evidence-based resource allocation, early warning capabilities, and policy evaluation. The model's strong performance on high-risk cases (96% F1-score) is particularly encouraging, as correctly identifying populations at greatest risk is crucial for effective intervention.

However, the models should be deployed with awareness of their limitations, including temporal validity concerns, ecological-level predictions, and the absence of certain important risk factors. These models are best viewed as decision-support tools that complement, rather than replace, expert judgment in public health planning.

In conclusion, this project demonstrates the valuable role that machine learning can play in addressing critical public health challenges. By combining computational power with domain knowledge and careful methodological design, we can develop tools that contribute meaningfully to UN Sustainable Development Goal 3 and ultimately help save lives through more effective suicide prevention strategies.

6. References

Dataset Source: [Kaggle - Suicide Rates Overview \(1985-2016\)](#)

Python Libraries: scikit-learn 1.0+, pandas, numpy, matplotlib, seaborn

World Health Organization (WHO) - Suicide Data Repository

United Nations Sustainable Development Goals - Goal 3: Good Health and Well-being

Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. JMLR 12, pp. 2825-2830

Github : https://github.com/prayan-gits/2462985_PrayanPiya_AI/tree/main/Assignments/Assignment%203

7.Appendix

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